507 Final Solve 2019

VU Zoologists



VU Zoologists Group

Efforts From All Admins

ارادہ پختہ ہو یقین کامل ہو زندگی میں
منزل ایک ہو تو نا ممکن کچھ نہیں ہوتا

عامر انظرؔ

Best Of Luck To All Fellows

Final 2019 Current Papers
Solve Objectives +Subjective

VU Zoologists

All Efforts By Admin:
Amir Ahmadani

"تم میں سب سے بہتر وہ ہے جو کسی کے کام آئے"



واللہ باخود آپ مجھ سے پرہیز کیجیے
میں بندہ ناچیز ہوں بڑے کام کی چیز
عامر انظرؔ

دعاوں میں یاد ضرور رکھیے گا

DEMO: Khalid Javed

Time Left 60 sec(s)

ZOO507:Quiz 2

Quiz Start Time: 12:00 AM, 31 July 2019

Question # 9 of 10 (Start time: 12:02:49 AM, 31 July 2019)

Total Marks: 1

What is Lichens?

Select the correct option

- Association between algae and fungus
- Association between animals and plants
- Association between algae and animals
- Association between plants and viruses

DEMO: Khalid Javed

Time Left 88 sec(s)

ZOO507:Quiz 2

Quiz Start Time: 12:00 AM, 31 July 2019

Question # 2 of 10 (Start time: 12:00:32 AM, 31 July 2019)

Total Marks: 1

carbon dioxide gas, makes up.....of the volume of the atmosphere and is also dissolved in water?

Select the correct option

- 0.138%
- 0.039%
- 0.037%
- 0.038% 

DEMO: Khalid Javed

Time Left 87 sec(s)

Z00507:Quiz 2

Quiz Start Time: 12:00 AM, 31 July 2019

Question # 8 of 10 (Start time: 12:02:35 AM, 31 July 2019)

Total Marks: 1

.....is a kind of coloration or patterns that help an animal to appear to blend with its surroundings.

Select the correct option

- Camouflage
- Crypsis
- Esophage
- Bacteriophage

DEMO: Khalid Javed

Time Left 87 sec(s)

Z00507:Quiz 2

Quiz Start Time: 12:00 AM, 31 July 2019

Question # 1 of 10 (Start time: 12:00:09 AM, 31 July 2019)

Total Marks: 1

Commensalism is a _____ relationship?

Select the correct option

- Harmful for both species
- Useful for both species
- Useful for one species and harmful for other
- Useful for one species and second is neither helped nor harmed

DEMO: Khalid Javed

Time Left 88 sec(s)

ZOO507:Quiz 2

Quiz Start Time: 12:00 AM, 31 July 2019

Question # 7 of 10 (Start time: 12:01:51 AM, 31 July 2019)

Total Marks: 1

The process that involves the cyclic movement of nutrients between biotic and abiotic environment of earth is termed as .....?

Select the correct option

- geochemical cycle
- Biogeochemical cycle
- Oxygen cycle
- Water cycle

DEMO: Khalid Javed

Time Left 87 sec(s)

ZOO507:Quiz 2

Quiz Start Time: 12:00 AM, 31 July 2019

Question # 3 of 10 (Start time: 12:00:50 AM, 31 July 2019)

Total Marks: 1

pollutants come from two types of sources?

Select the correct option

- Point sources and Non point sources
- Point sources and point mutation
- source point
- sources and Non point sources

DEMO: Khalid Javed

Time Left

ZOO507 Quiz 2

Quiz Start Time: 12:00 AM, 31 J

Question # 10 of 10 (Start time: 12:03:13 AM, 31 July 2019)

Total

..... is a measure of the evenness in distribution of the individual organisms among the species present.

Select the correct option

- dominance
- Succession
- diversity
- Coevolution

DEMO: Khalid Javed

Time Left 88 sec(s)

Z00587:Quiz 2

Quiz Start Time: 12:00 AM, 31 July 201

Question # 6 of 10 (Start time: 12:01:39 AM, 31 July 2019)

Total Marks:

water cycle involves ..... major processes?

Select the correct option

- Three and a half
- Two
- Four
- Three

DEMO: Khalid Javed

Time Left 86 sec(s)

ZOO507:Quiz 2

Quiz Start Time: 12:00 AM, 31 July 2019

Question # 5 of 10 (Start time: 12:01:24 AM, 31 July 2019)

Total Marks: 1

.....is a common strategy used to escape from predators.

Select the correct option

- oleander plants
- Foul tasting
- stinging nettles
- Chemical warfare

DEMO: Khalid Javed

Time Left 86 sec(s)

ZOO507:Quiz 2

Quiz Start Time: 12:00 AM, 31 July 2019

Question # 5 of 10 (Start time: 12:01:24 AM, 31 July 2019)

Total Marks: 1

.....is a common strategy used to escape from predators.

Select the correct option

- oleander plants
- Foul tasting
- stinging nettles
- Chemical warfare

DEMO: Khalid Javed

Time Left

ZOO567 Quiz 2

Quiz Start Time: 12:00 AM, 31 J

Question # 10 of 10 (Start time: 12:03:13 AM, 31 July 2019)

Total

..... is a measure of the evenness in distribution of the individual organisms among the species present.

Select the correct option

- dominance
- Succession
- diversity
- Coevolution

DEMO: Khalid Javed

Time Left 87 sec(s)

ZOO507:Quiz 2

Quiz Start Time: 12:00 AM, 31 July 2019

Question # 3 of 10 (Start time: 12:00:50 AM, 31 July 2019)

Total Marks: 1

pollutants come from two types of sources?

Select the correct option

- Point sources and Non point sources
- Point sources and point mutation
- source point
- sources and Non point sources

Question No : 1 of 26 Marks: 1 (Budgeted Time 1 Min)

Environmental science is an interdisciplinary study of connections between the earth's life-support system and _____

Answer (Please select your correct option)

- Human activities
- Human life
- Human behaviour
- All given

Question No : 2 of 26 Marks: 1 (Budgeted Time 1 Min)

Which one of the following falls under the category of renewable resources?

Answer (Please select your correct option)

- Copper
- Oil
- Coal
- Soil

Question No : 3 of 26 Marks: 1 (Budgeted Time 1 Min)

Economic growth is an increase in a nation's _____of goods and services

Answer (Please select your correct option)

- Output
- Input
- GDP
- All given

DEMO: Khalid Javed

Time Left 88 sec(s)

ZOO507:Quiz 2

Quiz Start Time: 12:00 AM, 31 July 2019

Question # 4 of 10 (Start time: 12:01:09 AM, 31 July 2019)

Total Marks: 1

A mutualistic association between algae and fungus in which both the partners get benefits from each other is known as

Select the correct option

- lichen
- Crustose
- Parasitism
- Ascomycetes

Question No : 4 of 26 Marks: 1 (Budgeted Time 1 Min)

Solar energy is called a _____ because it is renewed continuously and is expected to last at least 6 billion years as the sun completes its life cycle

Answer (Please select your correct option)

- Renewable resource
- Direct source
- Perpetual resource
- All given

Question No : 5 of 26 Marks: 1 (Budgeted Time 1 Min)

Some nonrenewable resources, such as _____ and aluminum, can be recycled or reused to extend supplies

Answer (Please select your correct option)

- Copper
- Sodium
- Iron
- Calcium

Question No : 6 of 26 Marks: 1 (Budgeted Time 1 Min)

A positive feedback loop causes a system to change further in the _____direction from which it is moving

Answer (Please select your correct option)

- Same
- Opposite
- Side by side
- Prallel

Viola tricolor lives in _____ environment

Answer (Please select your correct option)

- Acidic
- Basic
- Neutral
- Alkaline

Question No : 8 of 26 Marks: 1 (Budgeted Time 1 Min)

Cockroaches are present on earth since _____ million years.

Answer (Please select your correct option)

- 350
- 440
- 468
- 237

Question No : 9 of 26 Marks: 1 (Budgeted Time 1 Min)

Non native species that have a potential to cause economic or environmental harm to humans are termed as _____ species

Answer (Please select your correct option)

- Invasive
- Specialist
- Generalist
- Endemic

A certain range of values within the tolerance range is called the range of _____

Answer (Please select your correct option)

- Minimum
- Maximum
- Optimum
- All given

Question No : 11 of 26 Marks: 1 (Budgeted Time 1 Min)

Algae lives in _____ type of ecosystem

Answer (Please select your correct option)

- Land
- Forest
- Aquatic
- All given

Question No : 12 of 26 Marks: 1 (Budgeted Time 1 Min)

_____is a process that starts an ecological food web

Answer (Please select your correct option)

- Respiration
- Photosynthesis
- Reproduction
- Decomposition

Fungi belongs to _____ class of organisims

Answer (Please select your correct option)

- Producers
- Primary consumers
- Decomposers
- Detrivores

Question No : 14 of 26 Marks: 1 (Budgeted Time 1 Min)

Animal populations change over time as a result of_____

Answer (Please select your correct option)

- Birth
- Death
- Dispersal
- All given

Question No : 15 of 26 Marks: 1 (Budgeted Time 1 Min)

Which factor is helpful for the regulation of population?

Answer (Please select your correct option)

- Death rate
- Birth rate
- Climate
- All given

About _____of earth surface is covered with water.

Answer (Please select your correct option)

- 71%
- 56 %
- 91 %
- 55 %

Question No : 17 of 26 Marks: 1 (Budgeted Time 1 Min)

_____ is a process of decomposition and disintegration that causes the rocks to break down in a natural way

Answer (Please select your correct option)

- Weathering
- Migration
- Destruction
- Decomposition

Question No : 18 of 26 Marks: 1 (Budgeted Time 1 Min)

_____ is the example of inland wetland

Answer (Please select your correct option)

- Marches
- Swamps
- Flood plains
- All given

Question No : 19 of 26 Marks: 1 (Budgeted Time 1 Min)

Eutrophic lake usually have _____

Answer (Please select your correct option)

- Shallow depth
- Hihg turbidity
- High net primary net productivity
- All given

Question No : 20 of 26 Marks: 1 (Budgeted Time 1 Min)

The area of shoreline between low and high tides is called the _____

Answer (Please select your correct option)

- Tidal zone
- Intertidal zone
- Highplain zone
- Low level zone

ZOO-507

M.C.Qs

1) Endotherms produced their own body heat through..... Mechanism. Internal
2) Hibernating animals possess..... low metabolic rate
3) Human eye can detect electromagnetic radiations of wavelength range..... 380-760nm
4) .....is used an ecological indicators to evaluate natural biodiversity. Species composition
5) Tropical rainforest are the regions of species of..... more than 50%
6) The completion within the species..... Intraspecific
7) Foliose is an example of..... Lichens
8) .....is a kind of association in which one organisms called parasite lives on the expense of host. Parasitism
9) Some animals protect themselves by using prominent..... coloration
10) Grassland biome is an example of..... terrestrial biome
11) Sulfur circulates in..... Sulfur cycle
12) Chemically interactive nitrogen gas makes up..... Of the volume of atmosphere. 78%
13) El-nino effect which occurs every..... 2-7 years
14) A protective layer in the atmosphere that protects the life on earth from harmful effects of UV radiation from sun is..... ozone layer
15) Physical location of a gene..... locus
16) Use or defence of a resource by one individual that reduce the availability of resources to other individuals. competition
17) The term ecosystem coined by Tansley in..... 1935
18) Abiotic components includes..... Temperature
19) A community is made up of many..... Interact in an area. population
20) The ecological niche is an important concept of..... Structure. community
21) Cockroaches are present on earth since..... 350 million years
22) Invasive species is also called..... Alien species
23) A certain range of value within tolerance range is called the range of..... optimum
24) ..... Is a process that starts an ecological food web. photosynthesis
25) Spiders and insects are..... . (Secondary consumer)
26) Example of key stone species..... (bees, butterflies , hummingbirds, bats, alligator, wolf, leopard, lion, and
27) Food making processes called ..... (Photosynthesis)
28) Invasive species are also called..... (Alien Species)
29) Percentage of nitrogen in environment ..... (78%)
30) Hibernation is occur in large animal called .... (Winter Sleep)
31) Ozone is denoted as .... (O_3)
32) Mychriza is association between..... (Fungus & plant)
33) Living and non. Living things includes ----. (ecosystem)
34) Non. Renewable sources such as aluminium and ___. (copper)
35) comunity of different _ interact each other. (Population)
36) organisms are divided into __ types. (3)
37) viola tricolor live in _ environment (grasslands, wastelands, mainly on acidic or neutral soils. It is also found on the banks and in the alluviums)
38) ecological niche is important for the study of___. (Ecosystem)
39) blind indus dolphin belong to __ specie. (Endangered)
40) ____process start ecological food web. ()
41) ___Factor helpful to regulate the population? ()
42) ____%age of water that cover the earth surface. (71%)
43) __ zone contain mountain head water streams. (Transition)
44) Ectotherm organisms regulate their body temperature through__. (Environment)
45) CFCs stand for. (Cholroflouro carbons)
46) ___ is basic unit of life and responsible for the trait inherited. (Gene)
47) PAR stand for____. (Periodic Automatic Replenishment)
48) mRNA is furthur translated into ___. (Protein)
49) human affected environment __%. (40%)
50) sulphur circulate through ___ cycle. (Sulphur)
51) commensalism. ()

MCQs ZOO507 (Final)

1. Plants produce food using light through **PHOTOSYNTHESIS.**
2. Parasite and host in **PARASITISM.**
3. Animals escape in Aposematic through prominent **COLORATION.**
4. N2 is _____ in atmosphere. **78%**
5. Sulfur recycle through **SULFUR CYCLE.**
6. Excess nutrients cause high biological density **EUTROPHICATION.**
7. _____has living and nonliving. **ENVIRONMENT**
8. _____ disposal of waste. **IMPROPER**
9. Invasive species is called **ALIEN.**
10. Humans exploit _____ of biodiversity. **40%**
11. Industries and transportation produce **NOISE.**
12. mRNA translation makes **PROTEIN.**
13. Physical location of genes **LOCUS.**
14. _____ has ancestor and all descendants. **CLADE**
15. Food density and animal feeding rate is **FUNCTIONAL RESPONSE.**
16. Undesirable changes and contamination by **POLLUTANTS.**
17. Covers vast diversity area **GAMMA.**
18. Known species animals _____ plants _____. **70%, 22%**
19. El-Nino after **2-7 years.**
20. Grasslands are **TERRESTRIAL.**
21. Fungi & plants **MICORRHIZAE.**
22. Close relationship **SYMBIOSIS.**
23. Within species competition **INTRASPECIFIC.**
24. Organisms of same species compete **INTRASPECIFIC.**
25. Graphical representation of trophic levels **ECOLOGICAL PYRAMID.**
26. Sequence of energy flow through levels **FOOD CHAIN.**
27. Organism regulate body temperature **ENDOTHERNS.**
28. Beavers are ecological **ENGINEERS.**
29. during hibernation organism have **LOW METABOLISM.**
30. CO2 important for **ALL, GREEN HOUSE EFFECT, PHOTOSYNTHESIS & TRAP HEAT.**
31. No. of species **SPECIES RICHNESS.**
32. Succession on new area **PRIMARY SUCCESSION.**

Muhammad Zahid

33. Large original community **BIOME.**

34. Crust, sediments and rocks make **GEOSPHERE.**

35. Photosynthesis start **FOOD WEB.**

36. Ozone formula **O3.**

Short Questions:

Define Allele

Chemical warfare (3)

Exploitation. (2)

Hibernation. (2)

Keystone species (2)

Define mimicry (2)

Direct & Indirect exploitation. (3)

Mycorrhizae types. (3)

Industry ecology (3)

Primary and secondary Succession. (5)

Phosphorus cycle. (5)

Interspecific and intraspecific competition (5)

Global warming (10)

Effects of global warming (5)

Characters of community ecology

Type of nitrogen fixing bacteria

Cryptic coloration

Aposematic coloration with two examples

Source of water pollution

Biogeochemical cycle (10)

Greenhouse effect

Noise Pollution

Biodegradable and non-biodegradable

Growth of lichen and its factors

Bio magnification

Industrial ecology

Energy quality type

Biodiversity

Solid waste definition. (2)

Species richness (2)

El_nino effects (2)

Define symbiosis (2)

Alpha Diversity. (3)

Types of mycorrhizae (3)

Autogenic and allogeneic succession (3)

Main threats to biodiversity. (5)

Biogeochemical any two types (5)

Factors affecting genetic diversity. (10)

Counter shading

Urban and land ecology

Genetic diversity and its factors (10)

Source of energy

Water cycle

aestivation

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What ozone layer and its effect on human body

Commensalism and mutualism

Mimicry:

- Mimicry is the resemblance in external appearance, shape and colour between members of widely distinct families.
- The species which mimic is called **mimic** or **mimitec**.
- The species being intimated is known as **model**.
- In this organism wants to be seen.
- It helps an organism to gain fitness in environment.
- **Examples:** Koral snake and King snake, Viceroy butterfly and Monarch butterfly

solved

zoo-507 **Mcqs type questions for midtetm**

1) Living and non-living things includes **environment.**
2) Non re Newable resources such as aluminium and **copper.**
3) Community of different **population** interact with each other.
4) Organisms are divided into **2** types.
5) Viola tricolor live in **acidic** environment.
6) Ecological Niche is important for the study of **community structure.**
7) Blind indus dolphin belong to **platanista minor** species.
8) **Photosynthesis** Process start ecological web.
9) **78.09** % age of nitrogen in atmosphere ?
10) **Biotic and Abiotic Factor** helpful to regulate the population ?.
11) **71** % age of water that cover the earth surface.
12) **Source** Zone contain mountain head water stream.
13) Ecotherm organisms regulate their body temperature through **external environment.**
14) Wavelength of electromagnetic radiations detect human eye **380 to 740.**
15) Tropical rain forest contain **50%** of total world life.
16) **Interspecific** Competition among same species are called.
17) **Commensalism** is a relationship in which one member is benefited and the second neither helpful or harmful.
18) Foliose is growth form of **lichen.**
19) **Camouflage** it is kind of coloration or pattern that help an animal to appear to blend with surrounding.
20) Grass land biomes is example of **terrestrial biomes.**
21) E1 nino effect occurs in 2 to 7 years.
22) CFCs stands for **chlorofloro carbon.**
23) % age of plants **22 %** and animals **70 %** species on world.
24) **DNA** Is basic unit of life and responsible for the trait inherited.

VU Bio Mates

25) **Pollutants ?** anything in the environment that is harmful to health, sssss

26) Survival or activities of humans or other organisms.

27) **Interspecific** Competitions within a specie member.

28) Physical location of gene are **locus.**

29) mRNA is further translated into **protein.**

30) when extra nucleotide put into DNA is called **frameshift mutation.**

31) Invasive species are called **non native species.**

32) Hibernation in large animals known as **winter sleep.**

33) **Aposematic coloration** animals protected themselves by being dangerous or distasteful to pradators by using prominent coloration these are known as warning or aposematic coloration.

34) Sulphur circulate through **sulphur** cycle.

35) Actual place of an organism where it is lived is called **Habitat.**

36) Tiger belong to **tertiary** class of consumers.

37) An ecological pyramid is further classified into **3** main types.

38) **Food web** provide stability to ecosystem.

39) Cat fish is a **Detvirus.**

40) Grass hopper belong to class of consumer **primary.**

41) **Primary Consumer** eat producers first to get energy.

42) On land most producers are green plants which generally capture about **1 %** of solar energy.

43) Secondary consumers are also known as **meat eater.**

44) Atmosphere contain **21 %** of oxygen.

45) Study of human population is called **demography.**

46) A certain range of value with in tolerance range is called the range of **optimum.**

VU Bio Mates

Some mcqs becoz time was short

① Endotherm Produce Their own body heat through internal

(ii) ___ is the measure of no of specie Per unit area

(iii) In Aposmatic coloration animal Protect Themselves by using Prominent Coloration

(iv) Alga lives in aquatic type

(v) atmosphere contain 78% nitrogen

(vi) Eutrophic lake usually have Shallow depth

(vii) Aestivation is called

(viii) human eye can detect

(ix) Common strategy to escape from predator ___

(x) Term biodiversity was used by ___

(xi) Physical location of goal.

(xii) chemical inhibition of one species by another specie is ___

mcqs apnay liye check

3) animals ..... low metabolic rate
4) Human eye can detect electromagnetic radiations of wavelength range..... 380-760nm
5) .....is used an ecological indicators to evaluate natural biodiversity. Species composition
6) Tropical rainforest are the regions of species of..... more than 50%
7) The completion within the species..... Intraspecific
8) Follose is an example of..... Lichens
.....is a kind of association in which one organisms called parasite lives on the expense of host. Parasitism
9) Some animals protect themselves by using prominent..... coloration
10) Grassland biome is an example of..... terrestrial biome
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48) mRNA is furthur transfated into ___. (Protein)
49) human affected environment __ %. (40%)
50) sulphur circulate through ___ cycle. (Sulphur)
51) commensalism. ()

Sir Wattu 0335-4232497

1) ___ is the biological sc that studies how organisms of living things interact with their environ
(a) Biology (b) Ecology (c) geology (d) Immunology

2) Natural capital is supported by ___ source of energy
(a) Light (b) chemical (c) Heat (d) Electrical

3) Non renewable sources exist in a fixed quantity
(a) stock (b) Number (c) amount (d) all

4) The word ecology was first used by Ernest Haekel in ___ year
(a) 1857 (b) 1888 (c) 1988 (d) 1869

5) ___ is included in abiotic component of ecosystem
(a) carbon (b) Green plants (c) Temperature (d) Nitrogen

6) Autotrophs are further divided in to ___ sub classes
(a) 2 (b) 3 (c) 4 (d) 5

7) The harmful & toxic blooms of red ___ causes the colour of sea water to become red.
(a) Dinoflagellates (b) Diatoms (c) phytoplanktons (d) all

8) Atmosphere contains ___ % of nitrogen
(a) 21 (b) 78 (c) 35 (d) 87

9) What factor is helpful for regulation of population
(a) Death rate (b) birth rate (c) climate (d) all

10) Human population is increasing and present it is ___.
(a) 7.4 billion (b) 5 b (c) 8.1 b (d) 6.4 b

11) Geosphere include ___ components. (a) rock (b) soil (c) Sediment (d) all

12) ___ Zone contains mountain head water stream. (a) Source (b) Transition (c) Flood plain (d) all

13) Hibernating animals posses ___. (a) high & low metabolic (b) high (c) Low (d) high & low m.R

14) Human eye can detect electromagnetic radiation of wavelength of the range from ___
(a) 380–760nm (b) 379–760nm (c) 381–760nm (d) 380–761

15) ___ is used as an ecological indicator to evaluate natural biodiversity.
(a) Community Composition (b) ecological (c) biome (d) species

(16) Tropical rain forests are region in the world in which ___ % lime
(a) less than 50%. (b) more than 60%. (c) more than 50%. (d) 100%.

(17) The competition within the same species are known as
(a) intra specific (b) inter specific (c) inter & intra (d) inter & intra

(18) ___ is a relationship b/w two or more organism live closely together.
(a) chemical (b) biomes (c) close relation (d) symbiosis

(19) Foliose is the growth of ___ (a) lichens.

(20) The process that involves the cyclic movement b/w biotic & abiotic earth is termed as
(a) geochemical (b) Nitrogen (c) water (d) bio geochemical

MCQs ZOO507 (Final)

1. Plants produce food using light through	**PHOTOSYNTHESIS.**
2. Parasite and host in	**PARASITISM.**
3. Animals escape in Aposematic through prominent	**COLORATION.**
4. N2 is _____ in atmosphere.	**78%**
5. Sulfur recycle through	**SULFUR CYCLE.**
6. Excess nutrients cause high biological density	**EUTROPHICATION.**
7. _____has living and nonliving.	**ENVIRONMENT**
8. _____ disposal of waste.	**IMPROPER**
9. Invasive species is called	**ALIEN.**
10. Humans exploit _____ of biodiversity.	**40%**
11. Industries and transportation produce	**NOISE.**
12. mRNA translation makes	**PROTEIN.**
13. Physical location of genes	**LOCUS.**
14. _____ has ancestor and all descendants.	**CLADE**
15. Food density and animal feeding rate is	**FUNCTIONAL RESPONSE.**
16. Undesirable changes and contamination by	**POLLUTANTS.**
17. Covers vast diversity area	**GAMMA.**
18. Known species animals _____ plants _____.	**70%, 22%**
19. El-Nino after	**2-7 years.**
20. Grasslands are	**TERRESTRIAL.**
21. Fungi & plants	**MICORRHIZAE.**
22. Close relationship	**SYMBIOSIS.**
23. Within species competition	**INTRASPECIFIC.**
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25. Graphical representation of trophic levels	**ECOLOGICAL PYRAMID.**
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28. Beavers are ecological	**ENGINEERS.**
29. during hibernation organism have	**LOW METABOLISM.**
30. CO2 important for	**ALL, GREEN HOUSE EFFECT, PHOTOSYNTHESIS & TRAP HEAT.**
31. No. of species	**SPECIES RICHNESS.**
32. Succession on new area	**PRIMARY SUCCESSION.**

Muhammad Zahid

507

25/08/2019

Tehreem Fatima.

MCQ's (22) twice aya

1- Endotherms produced | Internal
2- competition with same device | Interspecific
3- Foliose is example | Lichen
4- El-nano effect | 2-7 years
5- physical location | Leaves
6- Sulfer circulated | Sulfer
7- oxygen coined | 1937
8- Percentage of nitrogen | 78% | twice (25)
9- Ozone emitted by | O_2
10- process of food web | photosynthesis (24)
11- Hybernation of large animal | winter sleep | twice
12- Invasive species are | alien
13- Abiotic component | Temperature
14- Mychorizea association | plant & fungi
15- county of different | Population (25)
16- MRNA further translated into | Protein | twice
17- PAR stands | PA sodiation
18- Ecological niche of Zoo | community
19- Hybernation animal | Low metabolic rate
20- Evaluate natural composition | Species composition
26- ____ process start ecological | Food web
27- CFC's stands for | chlorofluorocarbon
28- Food making process | Photosynthesis
29- disposal of waste | Improper
30- Insect & descendentents | CLADE
31- Geographical representation | Ecological pyramid
32- Excess nutrient | Eutrophication
33- Human exploits | 40%.
34- close relationship of both | Mutalism
35- Beavers are | Engineers
36- succession on new area | Primary
37-
38- CO_2 plays important role | Green house effect
39-
40- Deforestation decreases the carbondioxide in ecosyste

ZOO-507

M.C.Qs

1) Endotherms produced their own body heat through..... Mechanism. Internal
2) Hibernating animals possess..... low metabolic rate
3) Human eye can detect electromagnetic radiations of wavelength range..... 380-760nm
4) .....is used an ecological indicators to evaluate natural biodiversity. Species composition
5) Tropical rainforest are the regions of species of..... more than 50%
6) The completion within the species..... Intraspecific
7) Foliose is an example of..... Lichens
8) .....is a kind of association in which one organisms called parasite lives on the expense of host. Parasitism
9) Some animals protect themselves by using prominent..... coloration
10) Grassland biome is an example of..... terrestrial biome
11) Sulfur circulates in..... Sulfur cycle
12) Chemically interactive nitrogen gas makes up..... Of the volume of atmosphere. 78%
13) El-nino effect which occurs every..... 2-7 years
14) A protective layer in the atmosphere that protects the life on earth from harmful effects of UV radiation from sun is..... ozone layer
15) Physical location of a gene..... locus
16) Use or defence of a resource by one individual that reduce the availability of resources to other individuals. competition
17) The term ecosystem coined by Tansley in..... 1935
18) Abiotic components includes..... Temperature
19) A community is made up of many..... Interact in an area. population
20) The ecological niche is an important concept of..... Structure. community
21) Cockroaches are present on earth since..... 350 million years
22) Invasive species is also called..... Alien species
23) A certain range of value within tolerance range is called the range of..... optimum
24) ..... Is a process that starts an ecological food web. photosynthesis
25) Spiders and insects are..... . (Secondary consumer)
26) Example of key stone species..... (bees, butterflies , hummingbirds, bats, alligator, wolf, leopard, lion, and
27) Food making processes called ..... (Photosynthesis)
28) Invasive species are also called..... (Alien Species)
29) Percentage of nitrogen in environment ..... (78%)
30) Hibernation is occur in large animal called .... (Winter Sleep)
31) Ozone is denoted as .... (O_3)
32) Mychriza is association between..... (Fungus & plant)
33) Living and non. Living things includes ----. (ecosystem)
34) Non. Renewable sources such as aluminium and ___. (copper)
35) comunity of different _ interact each other. (Population)
36) organisms are divided into __ types. (3)
37) viola tricolor live in _ environment (grasslands, wastelands, mainly on acidic or neutral soils is also found on the banks and in the alluviums)
38) ecological niche is important for the study of___. (Ecosystem)
39) blind indus dolphin belong to __ specie. (Endangered)
40) ____process start ecological food web. ()
41) ___Factor helpful to regulate the population? ()
42) ____%age of water that cover the earth surface. (71%)
43) __ zone contain mountain head water streams. (Transition)
44) Ectotherm organisms regulate their body temperature through__. (Environment)
45) CFCs stand for. (Cholroflouro carbons)
46) ___ is basic unit of life and responsible for the trait inherited. (Gene)
47) PAR stand for____. (Periodic Automatic Replenishment)
48) mRNA is furthur translated into ___. (Protein)
49) human affected environment __ %. (40%)
50) sulphur circulate through ___ cycle. (Sulphur)
51) commensalism. ()

MCQs ZOO507 Final 24-8-2019 11AM.
1. Plants produce food using light through PHOTOSYNTHESIS. 2. Parasite and host in PARASITISM. 3. Animals escape in Aposematic through prominent COLORATION. 4. N2 is 78%in atmosphere. 5. Sulfur recycle through SULFUR CYCLE. 6. Excess nutrients cause high biological density EUTROPHICATION. 7. ENVIRONMENT has living and non living. 8. IMPROPER disposal of waste. 9. Invasive species is called ALIEN. 10. Humans exploit 40% of biodiversity. 11. Industries and transportation produce NOISE. 12. mRNA translation makes PROTEIN. 13. Physical location of genes LOCUS. 14. CLADE has ancestor and all descendants. 15. Food density and animal feeding rate is FUNCTIONAL RESPONSE. 16. Undesirable changes and contamination by POLLUTANTS. 17. Covers vast diversity

FUNCTIONAL RESPONSE. 16. Undesirable changes and contamination by POLLUTANTS. 17. Covers vast diversity area GAMMA. 18. Known species animals 70% plants 22%. 19. El-Nino after 2-7 years. 20. Grasslands are TERRESTRIAL. 21. Fungi & plants MICORRHIZAE. 22. Close relationship SYMBIOSIS. 23. Within species competition INTRASPECIFIC. 24. Organisms of same species compete INTRASPECIFIC. 25. Graphical representation of trophic levels ECOLOGICAL **PYRAMID**. 26. **Sequence** of energy flow through levels FOOD CHAIN.
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(3)

~~~~ is a common stratgy used to escape from predator.

Oleandor Plant. Foul tasting.

Stinging nettles. Chemical, warfare ✓

~~~~ is a kind of coloration or pattern that help an animal to appear to blend with its enviroment.

Camuoflage ✓ Crypsis.

Esophage Bacterophage

ZOO507 QUIZ 2

*A mutalistic association between algae & fungus in which both the partener get benefit from each other is known.

Lichen ✓ Crustose

Parasitism Ascomycetes

is a measure of evenness in distrabution of the invidiual organism among the species present.

Dominance Succession

Diversity ✓ Coevolution

Carbon dioxide gas makes up of the volume of atmosphere and also dissolve in water.

0.138% 0.039% 0.037% 0.038% ✓

What is lichens.? ✓

Association between Algae & fungus

Association // Animal & Plant

// // Algae & Animal

// // plant & viruses.

Commensalism is a ____ relationship.
Harmfull for both species. Usefull 4 both species.
Useful for one species & Harmfull for other.
Usefull for one Species & second is neither helped nor harmed.

Pollutants came from two type of Sources.
Point Sources & non Point Sources.
Point Sources & point mutation
Source point, Sources & non, p-s.

The process that involve the cyclic movement of nutrients blw biotic & abiotic enviroment of earth is term ai.
Geochemical cycle Biogeochemical cycle.
Oxygen cycle water cycle

Water cycle involved in Three major step.
Three & a hedlf-, Two, four., Three

All current 2019 final papers of Zoo507 Subject
Some solve questions
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Today paper
Zoo 507
Time 11
Define explotation ?
Difference between direct and indirect explotation
Define mimicry
Define symbiosis
Counter shading
Global warming
Effects of global warming
Threats to biodiversity.

1. Aestivation 2 marks

Aestivation or summer sleep is a period of inactivity and low metabolic rate in animals in response to high temperature and dry season.

Aestivation is common in many invertebrates, reptiles, and amphibians. **Aestivation** is a kind of adaptation in some animals that helps them to survive in extended periods of drying. The animal usually enters a burrow as its environment begins to dry. It generally does not eat or drink and emerges again after moisture returns.

The Australian burrowing frog, *Cyclorana alboguttatus*, in its burrow and water-retaining skin.

2. Exploitation 2 marks

The diversity of interactions between herbivores and plants, between predators and prey, and between parasites, parasitoids, pathogens, and hosts can be grouped under the heading of exploitation--interactions between species that enhance the fitness of one individual at the expense of another.

3. Global warming 2 marks

The average temperature of earth surface is increasing day by day due to increase in concentration of greenhouse gases in the atmosphere. Human activities are main cause of it.

The earth is gradually heating up due to rise in temperature. This phenomenon **is called Global warming.** It is causing serious problems to the health of plants and animals and also the surrounding environment.

4.Counter shading 3 marks

Counter shading is a type of color pattern in animals in which there is a dark pigmentation on the upper surface of body and light pigmentation on the lower surface of body.

Counter shading helps an animal to blend in with its surrounding environment.

Counter shading is a kind of camouflage common in frog and toad eggs.

These eggs are darkly pigmented on top and lightly pigmented on the bottom. When a bird or other predator views the eggs from above, the dark of the top side hides the eggs from detection against the darkness below.

On the other hand, when fish view the eggs from below, the light undersurface blends with the bright air-water interface.

Penguins are another example that exhibit phenomenon of counter shading.

9.Resource depletion & biodiversity 5 marks

Humans are either *directly* or *indirectly* exploiting about 40% of the earth's net primary production.

It involves converting natural areas to agricultural uses, frequently substituting less efficient crop plants for native species.

The main threats to biological diversity arise from habitat destruction by expanding human populations. Habitat loss displaces thousands of native plants and animals. Some of the most important threatened natural areas include,

o Tropical rain forests **o Coastal wetlands** **o Coral reefs**

Of these, tropical rain forests have probably received the most attention.

Tropical rain forests cover only 7% of the earth's land surface, but they contain more than 50% of the world's species.

Tropical rain forests are being destroyed rapidly, mostly for agricultural production. About 76,000 Km^2 (an area larger than *Costa Rica*) is being cleared each year. Clearing of tropical rain forests achieves little, because the thin, nutrient-poor soils of tropical rain forests are exhausted within two years

6.Overexploitation 3 marks

Overexploitation of resources (or using resources at an unsustainable rate) is a critical problem in conservation.

The overexploitation or non-sustainable use of wildlife is closely linked and plays an increasing role in the loss of biodiversity

Overharvesting, non-sustainable use, and the illegal trade in some species are threatening not only their continued survival but also that of ecosystems and the livelihoods of communities and local economics that depend upon them.

Unsustainable hunting, fishing, logging, or gathering of wild populations leads to their commercial, ecological or global extinction.

Overexploitation can be divided into two major categories:

Direct exploitation ranges from commercial activities such as logging operations or trade in endangered species to subsistence hunting.

Indirect exploitation:

Indirect exploitation encompasses the unintentional mortality of non target species such as fish or turtles killed as by-catch in fishery operations. Both endanger species around the world.

27. Mimicery

Mimicry (L. mimus, to imitate) occurs when a species resembles one, or sometimes more than one, other species and gains protection by the resemblance.

These **six species of Heliconius** are all distasteful to bird predators. A bird that consumes any member of the six species is likely to avoid all six species in the future.

Some butterfly species, such as the non-poisonous viceroy, gain protection by looking and acting like the monarch, a protective device known as mimicry.

Viceroy butterfly mimics monarch butterfly. Some moths have wings that look like the eyes of much larger animals. Hind wings o Io moth resembles eyes of a much larger animal. When touched, snake caterpillar changes shape to look like head of snake

2:30 pm
Species richness 2marks
Keystone species 2
El_nino effects 2
Types of myccorrhiz 3
Global warming 3
Industry ecology 3
Autogenic and allogeneic 3
Biogeochemical any two types 5
Interspecific and intraspecific competition 5
Global warming 10.
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56.Species richness

Species richness is the measure of number of species per unit area found in a particular site or place.

Species richness is a common measure of variety of life on earth.

Species abundance

Species abundance is the number of individuals per species, and relative abundance refers to the evenness of distribution of individuals among species in a community.

Species evenness

Species evenness is a measure of the relative abundance of the different species that contribute to the richness of an area.

As species richness and evenness increase, species diversity also increases.

51.Keystone species 2

A **keystone species** is a species that has a disproportionately large effect on community structure relative to its biomass or abundance.

- **Keystone species** differ from foundation species in two main ways: they are more likely to belong to higher trophic levels (to be top predators), and they act in more diverse ways than foundation species, which tend to modify their environment.
- Different sources use different definitions for keystone and foundation species. In some sources, foundation species are viewed as a subcategory of keystone species.

52.Invasive species

Species whose populations have expanded dramatically and out-compete or displace native species, potentially threatening the structure and function of intact ecosystems.

When alien species are introduced unintentionally or deliberately for whatever purpose, some of them turn invasive, and cause decline or extinction of indigenous species.

The Nile perch introduced into Lake Victoria in east Africa led eventually to the extinction of an ecologically unique assemblage of more than 200 species of cichlid fish in the lake.

The recent illegal introduction of the African catfish (*Clarias gariepinus)* for aquaculture purposes is posing a threat to the indigenous catfishes in our rivers.

A well-known example is the brown tree snake, which was introduced inadvertently on many Pacific islands and subsequently caused the extinction of a number of native birds, bats and lizards.

There appear to be three major stages in the process of invasion by exotic species:

- **Dispersal**
- **Establishment**
- **Integration**

43.What is El-nino effect?

Another type of short-term climate change is the so-called *El Nino effect*, which occurs every 2 to 7 years around Christmas time, when the typical weather pattern in the Pacific Ocean breaks down.

The trade winds weaken, and low pressure establishes itself in the central Pacific. It causes the winds to blow into the Pacific Basin from the west. This change causes warm surface water to move toward South and Central America, reversing the directions of some currents and placing warm water along the coasts.

The warm water produces heavy rain over the coastal deserts, subjecting them to disastrous flooding and erosion. The western United States also experiences major flooding events as a result of El Nino.

1. This temporary influx of rain on arid and semiarid regions creates the conditions for disastrous down slope movements in the form of mud slides, destroying entire villages.
2. One result of this temporary climate change is the occurrence of major brush fires, which destroy not only vegetation but also wild and domestic animals
3. Shifts in warm water in the southern Pacific and Indian Oceans cause cooler water to move along the coast of Australia, resulting in decreased rainfall.
4. The warm water El Nino brings to the coasts of North and South America substantially reduces the upwelling of cold, deep water, adversely affecting the cold water fisheries.

mycorrhizae

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Mycorrhiza

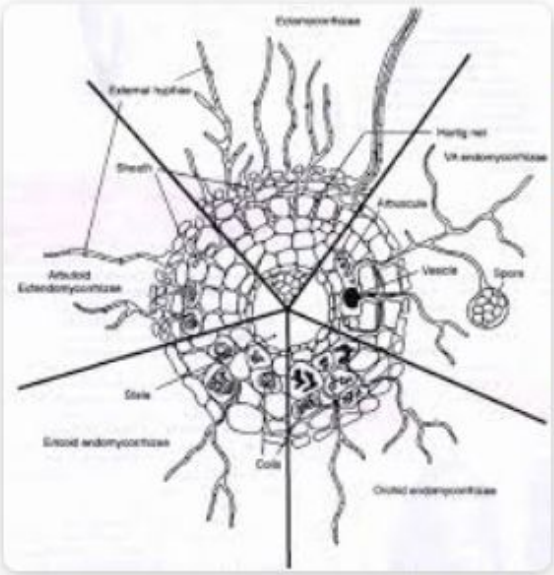


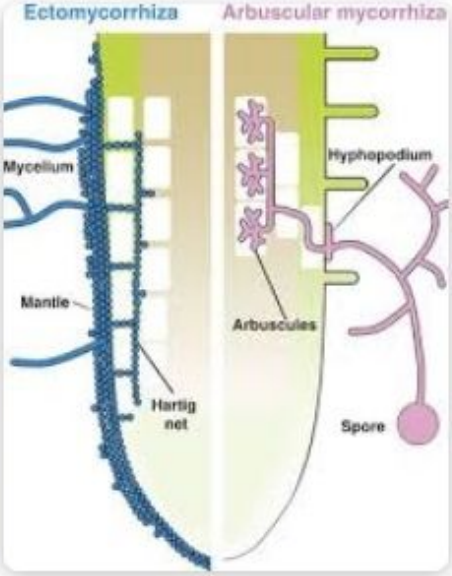


mycorrhizae, **mycorrhiza** or mycorrhizas) is a symbiotic association between a fungus and a plant. The term **mycorrhiza** refers to the role of the fungus in the plant's rhizosphere, its root system. **Mycorrhizae** play important roles in plant nutrition, soil biology and soil chemistry.

types mycorrhizae

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Types of Mycorrhizae

- Ectomycorrhiza. Ectomycorrhiza tend to form mutual symbiotic relationships with woody plants, including birch, beech, willow, pine, oak, spruce, and fir. ...
- Endomycorrhiza. ...
- Orchid **Mycorrhiza**. ...
- Arbuscular **Mycorrhiza**. ...
- Ericaceous **Mycorrhiza**. ...
- Arbutoid **Mycorrhiza**. ...
- Ectotrophic **Mycorrhiza**.

36.Difference of landscape ecology , and urban ecology

"**Landscape ecology** is concerned with spatial patterns in the landscape and how they develop, with an emphasis on the role of disturbance, including human impacts" (Smith and Smith).

- It is a relatively new branch of ecology, that employs Global Information Systems.

- The goal is to predict the responses of different organisms to changes in landscape, to ultimately facilitate ecosystem management.

Urban ecology

For ecologists, urban ecology is the study of ecology in urban areas, specifically the relationships, interactions, types and numbers of species found in urban habitats. Also, the design of sustainable cities, urban design programs that incorporate political, infrastructure and economic considerations.

37.industrial ecology

- The design of the **industrial infrastructure** such that it consists of a series of interlocking "technological ecosystems" interfacing with global natural ecosystems.

- **Industrial ecology** takes the pattern and processes of natural ecosystems as a design for sustainability. It represents a shift in paradigm from conquering nature to becoming nature.

18.Autogenic and allogenic succession.5

Autogenic succession

In autogenic succession the successional processes controlled by living or biotic component of an ecosystem

Allogenic succession

The entire successional process is controlled by non living or abiotic components of ecosystem.

Ecological succession is a process of gradual changes that causes the plants, animals, fungi and microorganism to become established in an area.

19.What is competition

Competition: use or defense of a resource by one individual that reduces the availability of the resource to other individuals

20.What is water cycle

The **hydrologic cycle, or water cycle**, collects, purifies, and distributes the earth's fixed supply of water.

Water is an amazing substance which makes the water cycle critical to life on earth.

The water cycle is a global cycle because there is a large reservoir of water in the atmosphere as well as in the hydrosphere, especially the oceans

What is a biochemical process?

Something that's **biochemical** relates to chemical **processes** that occur in living beings, like the chemical reactions in your body. ... **Biochemical processes** occur in living organisms. For example, the way your body converts food into energy is **biochemical**, as is the way your body fights disease and responds to drugs.

biochemical two types

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There are four **classes** of **biochemical** compounds: carbohydrates, proteins, lipids (fats), and nucleic acids.

What are two examples of biochemical?

Biochemical reactions are chemical **reactions** that take place inside living things. **Two** of the most important **biochemical reactions** are photosynthesis and cellular respiration. Photosynthesis is the process in which plants and certain other organisms synthesize glucose from carbon dioxide and water using light energy. Apr 20, 2012

28. Interaspecific competition and intraspecefic competition 5

Interspecific competition

All organisms require space and food for their survival. Requirement of both space and food causes competition between the species as well as among the same species.

The competition between **different** species for purpose of space and food is called **Interspecific competition.**

While competition within the **same** species is known as **Intraspecific competition.**

When members of different species compete for resources, one species may be forced to move or become extinct, or the two species may share the resource and coexist.

Most studies have shown that competing species can coexist. Coexistence can occur when species utilize resources in slightly different ways.

Robert MacArthur studied five species of warblers that all used the same caterpillar prey. Warblers partitioned their spruce tree habitats by dividing a tree into preferred regions for foraging. Although foraging regions overlapped, competition was limited, and the five species

29.green house effects (10)

Small amounts of certain gases, including water vapor (H_2O), carbon dioxide (CO_2), methane (CH_4), and nitrous oxide (N_2O), in the atmosphere play a role in determining the earth's average temperatures and its climates.

These **greenhouse gases** allow mostly visible light and some infrared radiation and ultraviolet (UV) radiation from the sun to pass through the atmosphere. The earth's surface absorbs much of this solar energy and transforms it to longer-wavelength infrared radiation (heat), which then rises into the lower atmosphere.

Some of this heat escapes into space, but some is absorbed by molecules of greenhouse gases and emitted into the lower atmosphere as even longer-wavelength infrared radiation.

Some of this released energy radiates into space, and some warms the lower atmosphere and the earth's surface. This natural warming effect of the troposphere **is called the greenhouse effect.**

Causes of Greenhouse effect

Human activities such as, Burning fossil fuel. Deforestation Growing crops release carbon dioxide, methane, and oxide into the atmosphere

10.Effects of global warming 10 marks

Some major effects are

Rise in temperature

- Rise in Sea Level
- Climate change
- Extinction
- Droughts
- Loss of biodiversity

Rise in Temperature:

The intense heat waves and rising temperatures are becoming more common as greenhouse gases are trapped in the atmosphere. The greenhouse effect thus leads to a rise in temperature, and as it becomes stronger, more heat is trapped within the planet.

Rise in Sea Level:

Warm surface temperatures cause glaciers, polar ice to melt. This in turn increases the amount of water in the world's oceans thus contributing to a rise in sea levels.

It threatens populations of coastal areas because of their higher vulnerability to flooding.

Climate change:

Gradual increase in temperature has also severely influenced the climate pattern of the earth surface. *El nino effect* is an example of changing climate due to Global warming.

Extinction:

The polar bear is considered to be an endangered species whose numbers are falling because of their inability to adapt to the volatile temperature changes in the Polar Regions.

Droughts:

A warmer climate will cause shortage of water supply and ultimately crop failure. If these water shortages are persistent it will cause a lot of disruptions in global food production by affecting agriculture and thus leading to situations such as starvation .

Loss of biodiversity:

Global warming causes drastic and irreversible changes both in the upper atmosphere and within the planet thus affecting it's every component including land, water, air etc.
It makes the organisms unable to survive in their native environments.

36.Difference of landscape ecology , and urban ecology

"**Landscape ecology** is concerned with spatial patterns in the landscape and how they develop, with a the role of disturbance, including human impacts" (Smith and Smith).

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507
24/8/19 time 5:00
Effect of global warming
Mimicry
Urban and land ecology
Characters of community ecology
Genetic diversity
Type of nitrogen fixing bacteria
Source of energy
Cryptis coloration
Water cycle
Keystone species.
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characteristics of community ecol

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Scholarly articles for **characteristics of community ecology**

Biological invasions: lessons for **ecology** - Lodge - Cited by 1944

... in structural and functional **characteristics** of an Indiana ... - Grossman - Cited by 457

What is microbial **community ecology**? - Konopka - Cited by 254

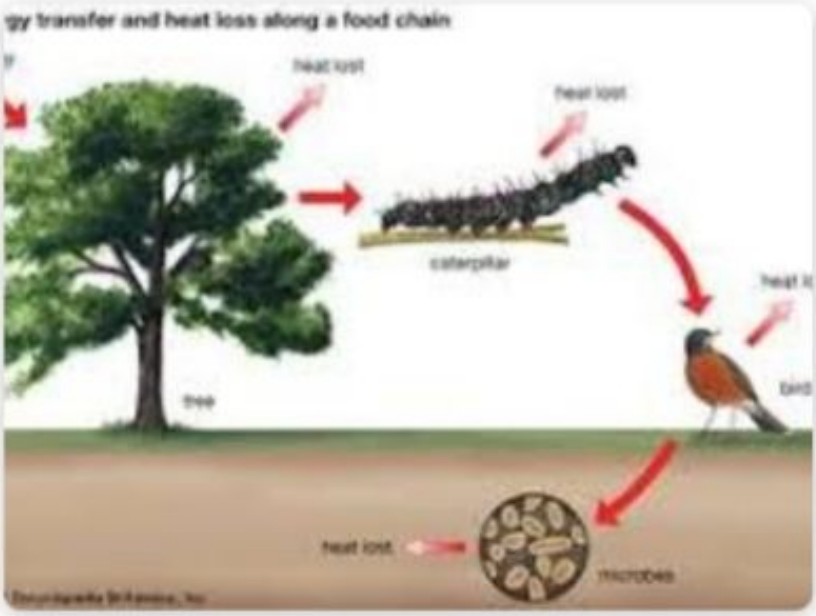
Some of the major **characteristics** of a **community ecology** are as follows: (a) Species Diversity (b) Growth From and structure (c) Dominance (d) Self reliance (e) Relative abundance (f) Trophic structure. **Community ecology** deals with the group of various kinds of population in the areas.

community ecology definition

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Community

Ecology





In ecology, a community is a group or association of populations of two or more different species occupying the same geographical area and in a particular time, also known as a biocoenosis. The term community has a variety of uses. Wikipedia

46.What are the causes of loss in genetic diversity ? 10 marks

Two ways of losing gene diversity in small populations,

- Changes in **the natural distribution** of genetic diversity among populations (artificial isolation and mixing)
- **Population size critical factor,**
- Census population size (Nc) vs. Effective population size (Ne), Usually Ne is much smaller that Nc (10 to 30%)

Factors responsible for genetic diversity loss

Four factors responsible for genetic diversity loss in small populations :

- Genetic drift
- Founder effect
- Demographic bottleneck
- Inbreeding

Prepared by: *Umar Ghafoor Wattu 0335-4232497*

Genetic drift

- " Random change in gene frequency within a population" (Meffe et al.),
- Not necessarily adaptive, Stronger in small populations

Founder effect

- "The establishment of a new population by a few original founders which carry only a small fraction of the total genetic variation of the parental population." (Ernst Mayr), Alleles may be lost, Differentiation from the parental population, **Example:** Irish Potato Famine

Demographic bottleneck

- Population suffers reduction in size and then recovers, Random losses of genetic diversity,
- Usually associated with catastrophic events or diseases,
- Genetic variation doesn't rebound from a decrease as quickly as population size.

Inbreeding

- *Inbreeding is breeding between close relatives*
- Leads to a reduction in genetic diversity (inbreeding depression: reduced health and fitness),
- The *primary problem* with inbreeding is that two closely related individuals are likely to have very similar genomes, and if one individual has a gene for a given negative trait, then the other is likely to have it as well., Inbreeding increases homozygosity (therefore decreases heterozygosity)

The other way to loss genetic diversity:

Changes in the natural distribution of genetic diversity among populations is related with the geographical distribution of the species and therefore with landscape management issues

Artificial isolation

Avoids genetic flux among populations due to barriers such as highways, dams, etc

Artificial mixing

Enhances genetic flux among populations where that flux was not possible due to natural barriers (examples: bridges, tunnels)

Allelic richness

Allelic richness Is the number of alleles in a sample (population),

Rare alleles

- are important during extreme environmental events,
- Loss of allelic richness is perhaps more serious than loss of quantitative variation because alleles are lost forever while quantitative variation can be recovered,
- Rare alleles are more important that their frequency in the population
- Example: Peppered moth and pollution in Manchester, Small sized populations are more prone to lose allelic richness

57.Types of nitrogen fixing bacteria

Azobactor

Free living aerobic bacteria

Clostridium

Free living anaerobic bacteria

Chlorobium

Photosynthetic bacteria

Thiobacillus

Chemosynthetic bacteria

Nostoc

Blue green algae

Prepared by: *Umar Ghafoor Wattu 0335-4232497*

A special vascular system develops in the host, supplying the products of photosynthesis to the nodule tissue and carrying away fixed nitrogen compounds to other parts of the plant. The nodule tissue provides shelter and carbon to bacteria.

14.Name three source of enegy.3

Organisms use one of three main sources of energy:

- **light**
- **inorganic molecules**
- **organic molecules**

15.Why species act as biological indicator.3

Because they evaluate the change in biodiversity.

32.Camouflage, biome, pollution (3 marks)

Camouflage is a kind of coloration or patterns that help an animal to appear to blend with its surroundings.

Many prey species use the camouflage (the ability to change color) of certain shapes or colors or (chameleons and cuttlefish).

Cryptic coloration (L. *crypticus, hidden)* *is a* type of camouflage that occurs when an animal takes on color patterns in its environment to prevent the animal from being seen by other animals.

The color pattern of this tiger (*Panthera tigris*) provides effective camouflage that helps when stalking prey. Praying mantises sit in flowers of a similar color and ambush visiting insects.

Some insect species have shapes that make them look like twigs ,bark, thorns, or even bird droppings on leaves. A leaf insect can be almost invisible against its background, as can an arctic hare in its white winter fur.

Biome A large regional community is known as biome. Each biome is differentiated from one another on the base of, Plants Animal Climate

33. Pollution

Pollution is anything in the environment that is harmful to the health, survival, or activities of humans or other organisms.

Pollution is any detrimental change in an ecosystem.

Most kinds of pollution are the results of human activities. Large human populations and demands for increasing goods and services contribute to pollution problems.

17.Three steps of water cycle.5

The **hydrologic cycle, or water cycle**, collects, purifies, and distributes the earth's fixed supply of water.

Water is an amazing substance which makes the water cycle critical to life on earth. The water cycle is a global cycle because there is a large reservoir of water in the atmosphere as well as in the hydrosphere, especially the oceans.

The water cycle is powered by energy from the sun and involves three major processes,

1. **Evaporation**

Prepared by: *Umar Ghafoor Wattu 0335-4232497*

2. **Transpiration**
3. **Precipitation**

Evaporation:

Evaporation changes liquid water into water vapor in the atmosphere. Incoming solar energy causes evaporation of water from the oceans, lakes, rivers, and soil.

About 84% of water vapor in the atmosphere comes from the oceans, which cover almost three- fourths of the earth's surface; the rest comes from land.

Transpiration:

Over land, about 90% of the water that reaches the atmosphere evaporates from the surfaces of plants through a process **called transpiration**.

Precipitation:

Precipitation returns the water back to earth's surface. Most precipitation becomes surface runoff that flows into streams and lakes and eventually in to the oceans. From oceans it can evaporate to repeat the water cycle.

Some precipitation is converted to ice that is stored. Some precipitation sinks to underground layers of rock, sand, and gravel called ***aquifers***, where it is stored as groundwater.

A small amount of the earth's water is absorbed by roots of plants. Some of this water is used to make food for consumers by photosynthesis, while most of water is evaporated through process of transpiration.

Consumers get their water from their food or by drinking it.

51.Keystone species 2

A **keystone species** is a species that has a disproportionately large effect on community structure relative to its biomass or abundance.

- **Keystone species** differ from foundation species in two main ways: they are more likely to belong to higher trophic levels (to be top predators), and they act in more diverse ways than foundation species, which tend to modify their environment.
- Different sources use different definitions for keystone and foundation species. In some sources, foundation species are viewed as a subcategory of keystone species.

52.Invasive species

Species whose populations have expanded dramatically and out-compete or displace native species, potentially threatening the structure and function of intact ecosystems.

When alien species are introduced unintentionally or deliberately for whatever purpose, some of them turn invasive, and cause decline or extinction of indigenous species.

The Nile perch introduced into Lake Victoria in east Africa led eventually to the extinction of an ecologically unique assemblage of more than 200 species of cichlid fish in the lake.

The recent illegal introduction of the African catfish (*Clarias gariepinus)* for aquaculture purposes is posing a threat to the indigenous catfishes in our rivers.

A well-known example is the brown tree snake, which was introduced inadvertently on many Pacific islands and subsequently caused the extinction of a number of native birds, bats and lizards.

There appear to be three major stages in the process of invasion by exotic species:

- **Dispersal**
- **Establishment**
- **Integration**

Toda paper zoo507
landscape and urbon ecology, biodegradable and non biodegradable, reforestation, genetic diversity, intraspecific and interspecific competition, autogenic and allogeneic succession, types of energy on the base of quality, biodiversity, industrial ecology, factor that effect the genetic diversity loss of low population.
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VU Zoologists

4. The warm water El Niño brings to the coasts of North and South America substantially reduces the upwelling of cold, deep water, adversely affecting the cold water fisheries.

44.Name two types of symbiosis. (2)

- **Commensalism**
- **Mutualism**

45.Difference between biodegradable and non degradable pollutants. (5)

- **Biodegradable pollutants** are harmful materials that can be broken down by natural processes. **Examples** are human sewage and newspapers.
- **Non-degradable pollutants** are harmful materials that natural processes cannot break down. **Examples** are toxic chemical elements such as lead, mercury, and arsenic

53.Reforestation: 2

Reforestation *is the process of restoring a forest that once existed but was removed at some point of time in the past.*

- It is basically a rehabilitation process of forests.
- Reforestation may occur naturally in a deforested area.
- However, we can speed it up by planting trees with due consideration to biodiversity that earlier existed in that area.

54.Strategy to Protect Ecosystems

Protecting and sustaining ecosystems and the biodiversity contained within them than on saving individual species is a burning issue in all over the world. Instead of preventing premature extinction of species, it is very important and interesting to protect threatened habitats and ecosystem services.

This ecosystems approach generally would employ the following four point plan:

- ❖ Map global ecosystems and create an inventory of the species contained in each of them and the ecosystem services they provide.
- ❖ Locate and protect the most endangered ecosystems and species, with emphasis on protecting plant biodiversity and ecosystem services.
- ❖ Seek to restore as many degraded ecosystems as possible.
- ❖ Make development biodiversity-friendly by providing significant financial incentives (such as tax breaks and write-offs) and technical help to private landowners who agree to help protect endangered ecosystems.

28. Interaspecific competition and intraspecefic competition 5

Interspecific competition

All organisms require space and food for their survival. Requirement of both space and food causes competition between the species as well as among the same species.

The competition between **different** species for purpose of space and food is called **Interspecific competition.**

While competition within the **same** species is known as **Intraspecific competition.**

When members of different species compete for resources, one species may be forced to move or become extinct, or the two species may share the resource and coexist.

Most studies have shown that competing species can coexist. Coexistence can occur when species utilize resources in slightly different ways.

Robert MacArthur studied five species of warblers that all used the same caterpillar prey. Warblers partitioned their spruce tree habitats by dividing a tree into preferred regions for foraging. Although foraging regions overlapped, competition was limited, and the five species

7. Primary succession with example 5 marks

The process by which a new community is propagated or initiated in an area where previously no life forms exist is **termed as primary succession**. Primary succession takes thousands of years to develop by the gradual accumulation of nutrients. Communities begin to establish in areas which are devoid of life.

The first community to become established in an area is called the **pioneer community.**

The organisms that develop this pioneer community are called **pioneers**.

Examples are lichens and mosses whose seeds or spores are distributed by the wind and carried by animals.

Death, decay, and additional nutrients add to the community. Over thousands of years nutrients accumulate, and the characteristics of the ecosystem changes.

Seral stage

Each successional stage is called a **seral stage.**

The entire successional sequence is termed as a **sere**.

Biomass increases, nutrients are conserved more efficiently, and productivity declines in the course of succession.

Climax community

The relatively stable and uniform community developed by the pioneers through different successional stages is named as **climax community**. It is a final and mature stage of any community.

Primary succession on a sand dune. Beach grass is the first species to become established. It stabilizes the dune so that shrubs, and eventually trees, can grow.

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CFCs are released by refrigerants. CFCs discharged in the lower part of atmosphere move upward and reach into stratosphere.

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Effects of Ozone depletion

- Skin cancer
- Snow-blindness (inflammation of cornea by UV-rays)
- Cataract
- Low crop production

18.Autogenic and allogenic succession.5

Autogenic succession

In autogenic succession the successional processes controlled by living or biotic component of an ecosystem

Allogenic succession

The entire successional process is controlled by non living or abiotic components of ecosystem.

Ecological succession is a process of gradual changes that causes the plants, animals, fungi and microorganism to become established in an area.

19.What is competition

Competition: use or defense of a resource by one individual that reduces the availability of the resource to other individuals

20.What is water cycle

The **hydrologic cycle, or water cycle**, collects, purifies, and distributes the earth's fixed supply of water.

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21.Define allele. 2 marks

- **Allele**: one of the genes at a particular locus.

13. Name of energy quality type.2

It determines the ability of an energy source to do useful work. Energy manifest itself in several forms such as heat, mechanical, electrical, magnetic, kinetic (in motion), potential, etc. Some types of energy are more useful than the others.

Two types of energy on the base of energy Quality,

1. **High quality energy**
2. **Low quality energy**

High quality energy

High-quality energy is concentrated form of energy and has a high capacity to do useful work. Examples are,

- Very high-temperature heat
- Nuclear fission
- Concentrated sunlight
- High-velocity wind
- Energy released by burning natural gas, gasoline, or coal.

Low quality energy

A dispersed form of energy that has a little capacity to do useful work.

Heat energy of moving water molecules is of low quality because it is dispersed in water molecules that lowers its temperature.

The total amount of heat stored in the Atlantic ocean is greater than the amount of high quality chemical energy stored in all the oil deposits of Saudi Arabia. But wide dispersion of this ocean's heat energy makes it unable to do useful work.

11. Biodiversity.2 marks

The **term biodiversity** was used by *Edward Wilson* to describe the combined diversity at all the levels of biological organization. **Biodiversity** is a vital renewable resource.

In our biosphere immense diversity or heterogeneity exists not only at the species level but at all levels of biological organization ranging from macromolecules within cells to biomes.

Biological diversity, or biodiversity, is the variety of all life forms on the earth.

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The variety of life on Earth at all its levels, from genes to ecosystems, and the ecological and evolutionary processes that sustain it.
Scientists have identified about 1.8 million species, and every year, thousands of new species are identified.
The identified species include almost a million species of insects, 270,000 plant species, and 45,000 vertebrate animal species.

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Zoo507 11:00 AM 27-08-2019
Hibernation 2m
Mimicry 2m
Genetic drift 2m
Industrial ecology 3m
Species hotspots 3m
Global warming 3m
Primary succesaion with example 5m
Urban and landscpe ecology 5m
Strategies to protect ecosystems 5m
Genectic diversity loss in small populations 10m
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50.Effects of Air pollution 5

- Acid rain
- Greenhouse effect
- Global warming
- Ozone layer destruction

Acid rain or deposition

Burning fossil fuels releases sulfur dioxide(SO_2) and nitrogen oxides(NO_x) into the atmosphere. Sulfur dioxide and water combine to produce sulfuric acid(H_2SO_4), which falls as acid deposition. Acid deposition lowers the pH of lakes.

Greenhouse effect

Carbon dioxide released from burning fuels accumulates in the atmosphere and prevent the escape of heat coming from earth. This effect is termed as Greenhouse effect.

Global warming

Carbon dioxide reflects solar radiation back to the earth. This reflection of solar radiation is predicted to cause an increase in world temperature, polar ice caps to melt, and ocean levels to rise.

Ozone layer destruction

The release of chlorinated fluorocarbons from aerosol cans, air conditioners, and refrigerators contributes to the depletion of ozone layer (the earth's ultraviolet filter). It will increase the chance of skin

58.Species hotspots

Species hotspots are geographic regions that support high level of species diversity.

Tropical rainforests are one of important species hotspots region in the world because they are home to more than **50% species**.+

59. Naturalism

Naturalism is the belief that nothing exists beyond the natural world. Instead of using supernatural or spiritual explanations, naturalism focuses on explanations that come from the laws of nature

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55.Major types of Mycorrhizae 3

1. Endomycorrhizas or Arbuscular mycorrhizas
2. Ectomycorrhizas
3. Ericoid mycorrhizas

46.What are the causes of loss in genetic diversity ? 10 marks

Two ways of losing gene diversity in small populations,

- Changes in **the natural distribution** of genetic diversity among populations (artificial isolation and mixing)
- **Population size critical factor,**
- Census population size (Nc) vs. Effective population size (Ne), Usually Ne is much smaller that Nc (10 to 30%)

Factors responsible for genetic diversity loss

Four factors responsible for genetic diversity loss in small populations :

- Genetic drift
- Founder effect
- Demographic bottleneck
- Inbreeding

Prepared by: *Umar Ghafoor Wattu 0335-4232497*

Genetic drift

- " Random change in gene frequency within a population" (Meffe et al.),
- Not necessarily adaptive, Stronger in small populations

Founder effect

- "The establishment of a new population by a few original founders which carry only a small fraction of the total genetic variation of the parental population." (Ernst Mayr), Alleles may be lost, Differentiation from the parental population, **Example:** Irish Potato Famine

Demographic bottleneck

- Population suffers reduction in size and then recovers, Random losses of genetic diversity,
- Usually associated with catastrophic events or diseases,
- Genetic variation doesn't rebound from a decrease as quickly as population size.

Inbreeding

- *Inbreeding is breeding between close relatives*
- Leads to a reduction in genetic diversity (inbreeding depression: reduced health and fitness),
- The *primary problem* with inbreeding is that two closely related individuals are likely to have very similar genomes, and if one individual has a gene for a given negative trait, then the other is likely to have it as well., Inbreeding increases homozygosity (therefore decreases heterozygosity)

The other way to loss genetic diversity:

Changes in the natural distribution of genetic diversity among populations is related with the geographical distribution of the species and therefore with landscape management issues

Artificial isolation

Avoids genetic flux among populations due to barriers such as highways, dams, etc

Artificial mixing

Enhances genetic flux among populations where that flux was not possible due to natural barriers (examples: bridges, tunnels)

Allelic richness

Allelic richness Is the number of alleles in a sample (population),

Rare alleles

- are important during extreme environmental events,
- Loss of allelic richness is perhaps more serious than loss of quantitative variation because alleles are lost forever while quantitative variation can be recovered,
- Rare alleles are more important that their frequency in the population
- Example: Peppered moth and pollution in Manchester, Small sized populations are more prone to lose allelic richness

Zoo507
Factor of biodiversity (5)
Factor of genetic biodiversity (10)
Define allele (2)
Define hibernation and winter sleep (3)
Define competition with exm (3)
Inter and intraspecific competition (2).
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40.Difference between hibernation and winter sleep

True hibernation occurs in small animals in which temperature substantially falls to remain alive in cold environment. Animals remain inactive for weeks or even for months.

Hibernation in large animals like bears is termed as winter sleep. Body temperature does not falls substantially and sleeping animals can wake and active quickly. It lasts for short period of time. Large animals have more energy reserves that enables them to survive in winter.

41.phosphorus cycle

Phosphorus circulates through water, the earth's crust, and living organisms in the phosphorus cycle.

In contrast to the cycles of water, carbon, and nitrogen, the phosphorus cycle does not include the atmosphere. The major reservoir for phosphorous is phosphate salts containing phosphate ions (PO_4^{-3}) in terrestrial rock formations and ocean bottom sediments. The phosphorus cycle is slow compared to the water, carbon, and nitrogen cycles.

Phosphorous is a component of biologically important molecules such as,

1. Nucleic acids
2. Energy transfer molecules (ADP and ATP)
3. Vertebrate bones and teeth.

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As water runs over exposed phosphorus containing rocks, it slowly erodes away inorganic compounds that contain phosphate ions (PO_4^{-3}). The dissolved phosphate can be absorbed by the roots of plants and by other producers.

Phosphate can be lost from the cycle when it washes from the land into streams and rivers and is carried to the ocean. In oceans it can be deposited as marine sediment and remain trapped for millions of years.

Someday, geological processes may uplift and expose these seafloor deposits, from which phosphate can be eroded to start the cycle again.

Because most soils contain little phosphate, it is often the limiting factor for plant growth on land unless phosphorus (as phosphate salts mined from the earth) is applied to the soil as an inorganic fertilizer

Today 29/8 11 Am
ZOO 507
ECOLOGY
*What is Allele.
*Define compitition and types intraspecefic and interspecefic compitution
*biodiversity
*genetic diversity
*ozone layer depletion+5
*global warming +10.



ecology definition

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DICTIONARY

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ecology

/ɪˈkɒlədʒi,ɛˈkɒlədʒi/

noun

the branch of biology that deals with the relations of organisms to one another and to their physical surroundings.

What is the simple definition of ecology?

Ecology is the science that studies the biota (living things), the environment, and their interactions. It comes from the Greek oikos = house; logos = study. **Ecology** is the study of ecosystems. Ecosystems describe the web or network of relations among organisms at different scales of organization.

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Effects of Ozone depletion

- Skin cancer
- Snow-blindness (inflammation of cornea by UV-rays)
- Cataract
- Low crop production

10.Effects of global warming 10 marks

Some major effects are

Rise in temperature

- Rise in Sea Level
- Climate change
- Extinction
- Droughts
- Loss of biodiversity

Rise in Temperature:

The intense heat waves and rising temperatures are becoming more common as greenhouse gases are trapped in the atmosphere. The greenhouse effect thus leads to a rise in temperature, and as it becomes stronger, more heat is trapped within the planet.

Rise in Sea Level:

Warm surface temperatures cause glaciers, polar ice to melt. This in turn increases the amount of water in the world's oceans thus contributing to a rise in sea levels.

It threatens populations of coastal areas because of their higher vulnerability to flooding.

Climate change:

Gradual increase in temperature has also severely influenced the climate pattern of the earth surface. *El nino effect* is an example of changing climate due to Global warming.

Extinction:

The polar bear is considered to be an endangered species whose numbers are falling because of their inability to adapt to the volatile temperature changes in the Polar Regions.

Droughts:

A warmer climate will cause shortage of water supply and ultimately crop failure. If these water shortages are persistent it will cause a lot of disruptions in global food production by affecting agriculture and thus leading to situations such as starvation .

Loss of biodiversity:

Global warming causes drastic and irreversible changes both in the upper atmosphere and within the planet thus affecting it's every component including land, water, air etc.
It makes the organisms unable to survive in their native environments.

Zoo 507
Time 11
Define explotation ?
Difference between direct and indirect explotation
Define mimicry
Define
Counter shading
Global warming
Effects of global warming
Threats to biodiversity.
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4.Counter shading 3 marks

Counter shading is a type of color pattern in animals in which there is a dark pigmentation on the upper surface of body and light pigmentation on the lower surface of body.

Counter shading helps an animal to blend in with its surrounding environment.

Counter shading is a kind of camouflage common in frog and toad eggs.

These eggs are darkly pigmented on top and lightly pigmented on the bottom. When a bird or other predator views the eggs from above, the dark of the top side hides the eggs from detection against the darkness below.

On the other hand, when fish view the eggs from below, the light undersurface blends with the bright air-water interface.

Penguins are another example that exhibit phenomenon of counter shading.

9.Resource depletion & biodiversity 5 marks

Humans are either *directly* or *indirectly* exploiting about 40% of the earth's net primary production.

It involves converting natural areas to agricultural uses, frequently substituting less efficient crop plants for native species.

The main threats to biological diversity arise from habitat destruction by expanding human populations. Habitat loss displaces thousands of native plants and animals. Some of the most important threatened natural areas include,

o Tropical rain forests **o Coastal wetlands** **o Coral reefs**

Of these, tropical rain forests have probably received the most attention.
Tropical rain forests cover only 7% of the earth's land surface, but they contain more than 50% of the world's species.

Tropical rain forests are being destroyed rapidly, mostly for agricultural production. About 76,000 Km^2 (an area larger than *Costa Rica*) is being cleared each year. Clearing of tropical rain forests achieves little, because the thin, nutrient-poor soils of tropical rain forests are exhausted within two years

507
Global warming 10 marks
Chemical warfare
Direct or indirect explotion
Species richness
Autogenic allogenic succession
Intraspcific interspecific competition
Exploition.
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25.Chemical warfare

Chemical warfare is another common strategy to escape from predators.

Chemical warfare refers to use of chemical substance as a weapon for defence purpose.

Some predators use chemical warfare to attack their prey. **For example**, spiders and poisonous snakes use venom to paralyze their prey and to deter their predators.

Some prey species discourage predators with chemicals that are poisonous (oleander plants)and irritating (stinging nettles and bombardier beetles).

Bombardier beetle

Some possess foul smelling (skunks, skunk cabbages, and stinkbugs), or bad tasting (buttercups and monarch butterflies). Foul tasting monarch butterfly

When attacked, some species of squid and octopus emit clouds of black ink, allowing them to escape by confusing their predators.

56.Species richness

Species richness is the measure of number of species per unit area found in a particular site or place.

Species richness is a common measure of variety of life on earth.

Species abundance

Species abundance is the number of individuals per species, and relative abundance refers to the evenness of distribution of individuals among species in a community.

Species evenness

Species evenness is a measure of the relative abundance of the different species that contribute to the richness of an area.

As species richness and evenness increase, species diversity also increases.

Global warming 10m
Aestivation
Industrial ecology
Genetic draft
Mutualism n commensalism
Primary succession.

Admin
Amir Ahmadani
VU Zoologists
Best of luck to you all

1. Aestivation 2 marks

Aestivation or summer sleep is a period of inactivity and low metabolic rate in animals in response to high temperature and dry season.

Aestivation is common in many invertebrates, reptiles, and amphibians. **Aestivation** is a kind of adaptation in some animals that helps them to survive in extended periods of drying. The animal usually enters a burrow as its environment begins to dry. It generally does not eat or drink and emerges again after moisture returns.

The Australian burrowing frog, *Cyclorana alboguttatus*, in its burrow and water-retaining skin.

22.Define Industrial ecology. 2 marks

- The design of the **industrial infrastructure** such that it consists of a series of interlocking *"technological ecosystems"* interfacing with global natural ecosystems.
- **Industrial ecology** takes the pattern and processes of natural ecosystems as a design for sustainability. It represents a shift in paradigm from conquering nature to becoming nature.

23.Factors of water cycle

The **hydrologic cycle, or water cycle**, collects, purifies, and distributes the earth's fixed supply of water.

Water is an amazing substance which makes the water cycle critical to life on earth. The water cycle is cycle because there is a large reservoir of water in the atmosphere as well as in the hydrosphere, esp oceans.

26.Commensalism or mutualism (5)

Commensalism is a symbiotic relationship in which one member of the relationship benefits, and the second is neither helped nor harmed.

In tropical forests certain kinds of silverfish insects move along with columns of army ants to share the food obtained by the ants in their raids. The army ants receive no apparent harm or benefit from the silverfish.

Epiphytic plants such as certain types of orchids and bromeliads attach themselves to the trunks or branches of large trees in tropical and subtropical forests. These air plants benefit by getting a solid base on which they grow.

They also live in an elevated spot that gives them better access to sunlight, water from the humid air and rain, and nutrients falling from the tree's upper leaves and limbs. Their presence apparently does not harm the tree.

Mutualism

Mutualism is a symbiotic relationship that benefits both members. In mutualism, two species behave in ways that benefit both by providing each with food, shelter, or some other resource.

1. **For example,** honeybees, caterpillars, butterflies, and other insects feed on a male flower's nectar, picking up pollen in the process, and then pollinating female flowers when they feed on them.
2. In gut inhabitant mutualism, vast armies of bacteria in the digestive systems of animals help to break down (digest) their hosts' food. In turn, the bacteria receive a sheltered habitat and food from their host.
3. Oxpeckers or tickbirds feed on parasitic ticks that infest large, thick skinned animals such as the endangered black rhinoceros.

Prepared by: *Umar Ghafoor Wattu 0335-4232497*

4. A clownfish gains protection and food by living among deadly stinging sea anemones and helps protect the anemones from some of their predators.

Some other forms of mutualism are,

- **Lichen (**Algea & Fungus**)**
- **Mycorrhizae** (Algea & Plant)
- **Root nodules**

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Efforts From All Admins

ارادہ پختہ ہو یقین کامل ہو زندگی میں
منزل ایک ہو تو نا ممکن کچھ نہیں ہوتا

عامر انظرؔ

Best Of Luck To All Fellows

Final 2019 Current Papers
Solve Objectives +Subjective

VU Zoologists

All Efforts By Admin:
Amir Ahmadani

"تم میں سب سے بہتر وہ ہے جو کسی کے کام آئے"



واللہ باخود آپ مجھ سے پرہیز کیجیے
میں بندہ ناچیز ہوں بڑے کام کی چیز
عامر انظرؔ

دعاوں میں یاد ضرور رکھیے گا